

Pentose Phosphate Pathway

Also known as:

Pentose shunt

Hexose monophosphate shunt

Phosphogluconate pathway

Dr Balakrishnan. S, PhD.,
Senior Lecturer of Biochemistry

Objectives:

- At the end of this lecture, it is expected to;
 - State the Biomedical Importance of Hexose monophosphate (HMP) shunt
 - Describe the reactions of HMP shunt in details
 - Understand the differences and connections between HMP shunt and glycolysis
 - Understand the regulations of HMP shunt
 - Describe the consequences of HMP shunt enzymes deficiencies:
 - Glucose 6-Phosphate Dehydrogenase Deficiency
 - FAVISM
 - Thiamine deficiency and transketolase activity

Moderate glucose flux

**Glycolysis
only** →



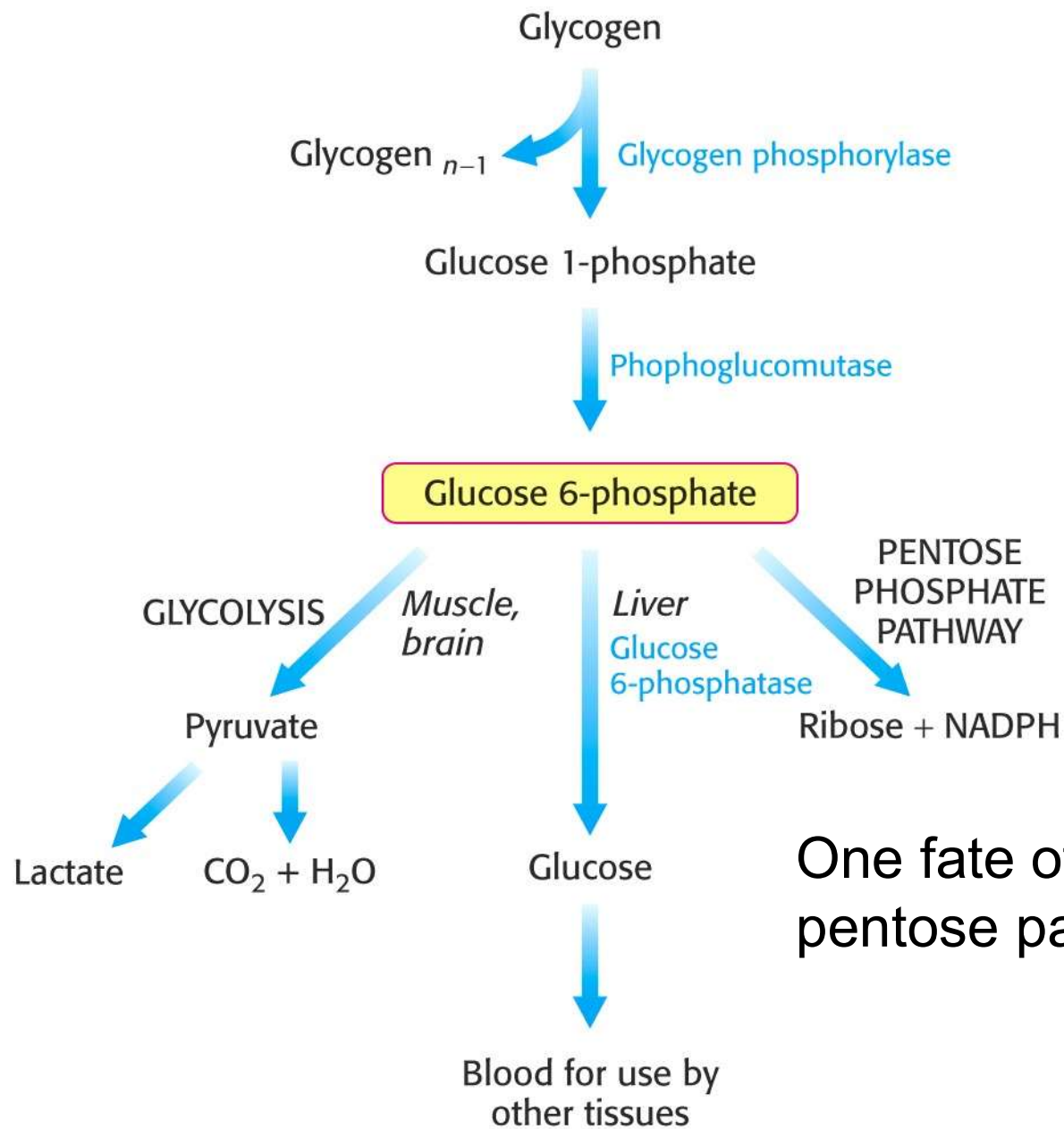
Large glucose flux

Glycolysis



**Pentose
Phosphate
Pathway**





One fate of G6P is the pentose pathway.

Biomedical Importance:

- An alternative route for the metabolism of glucose.
- Does not generate ATP but has two major functions:
 - (1) Formation of **NADPH** for synthesis of fatty acids and steroids.
 - (2) Synthesis of **ribose** for nucleotide and nucleic acid formation.
- Genetic deficiency of glucose 6-phosphate dehydrogenase cause **hemolytic anemia**
 - Affecting approx. 100 million people worldwide.

Demand for NADPH:

- Biosynthetic pathways
 - FA synthesis (liver, adipose, mammary)
 - Cholesterol synthesis (liver)
 - Steroid hormone synthesis (adrenal, ovaries, testes)
- Detoxification (Cytochrome P-450 System) – liver
- Formation of Reduced glutathione as an antioxidant (RBC)

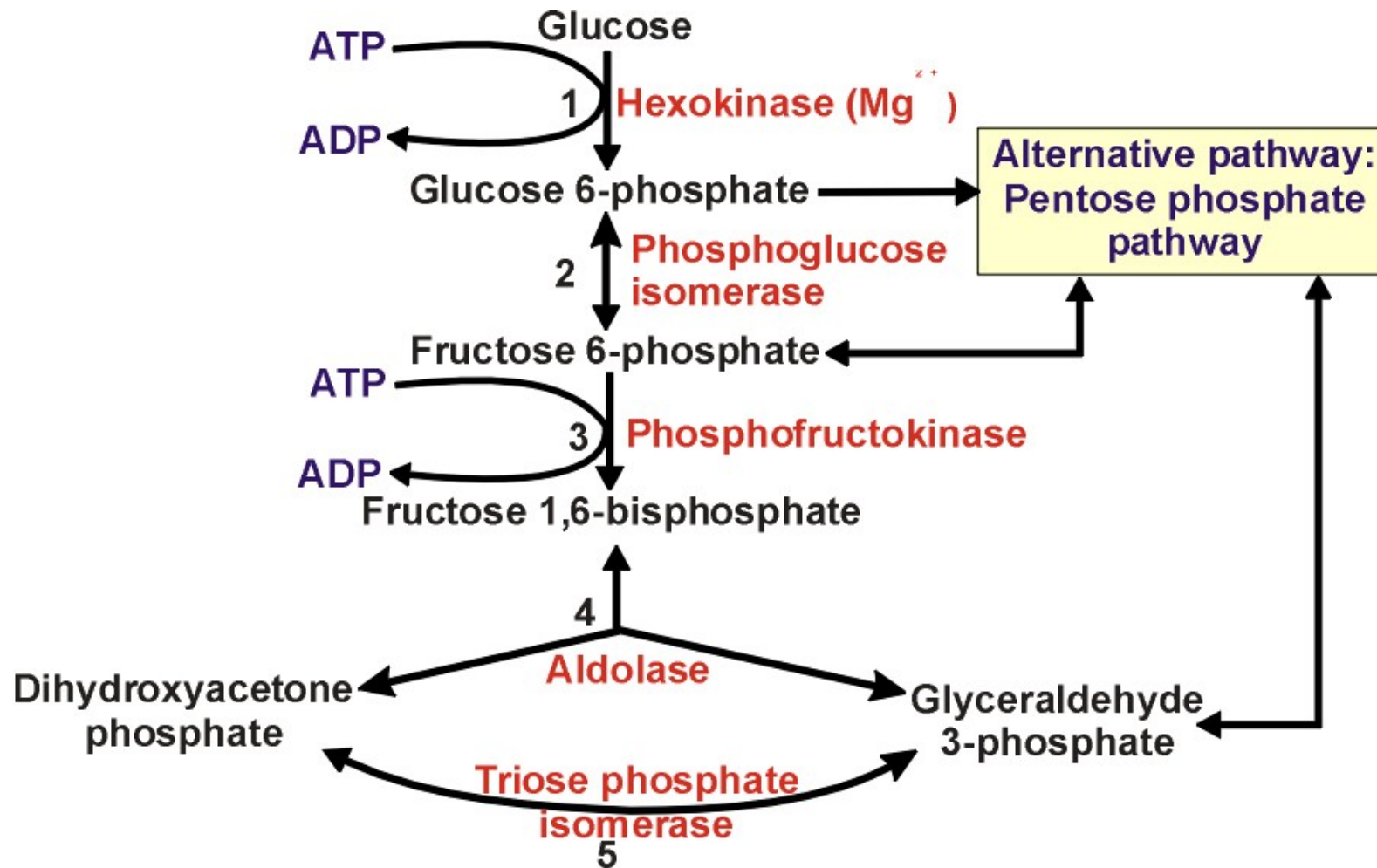
TABLE 20.4 Tissues with active pentose phosphate pathways

Tissue	Function
Adrenal gland	Steroid synthesis
Liver	Fatty acid and cholesterol synthesis
Testes	Steroid synthesis
Adipose tissue	Fatty acid synthesis
Ovary	Steroid synthesis
Mammary gland	Fatty acid synthesis
Red blood cells	Maintenance of reduced glutathione

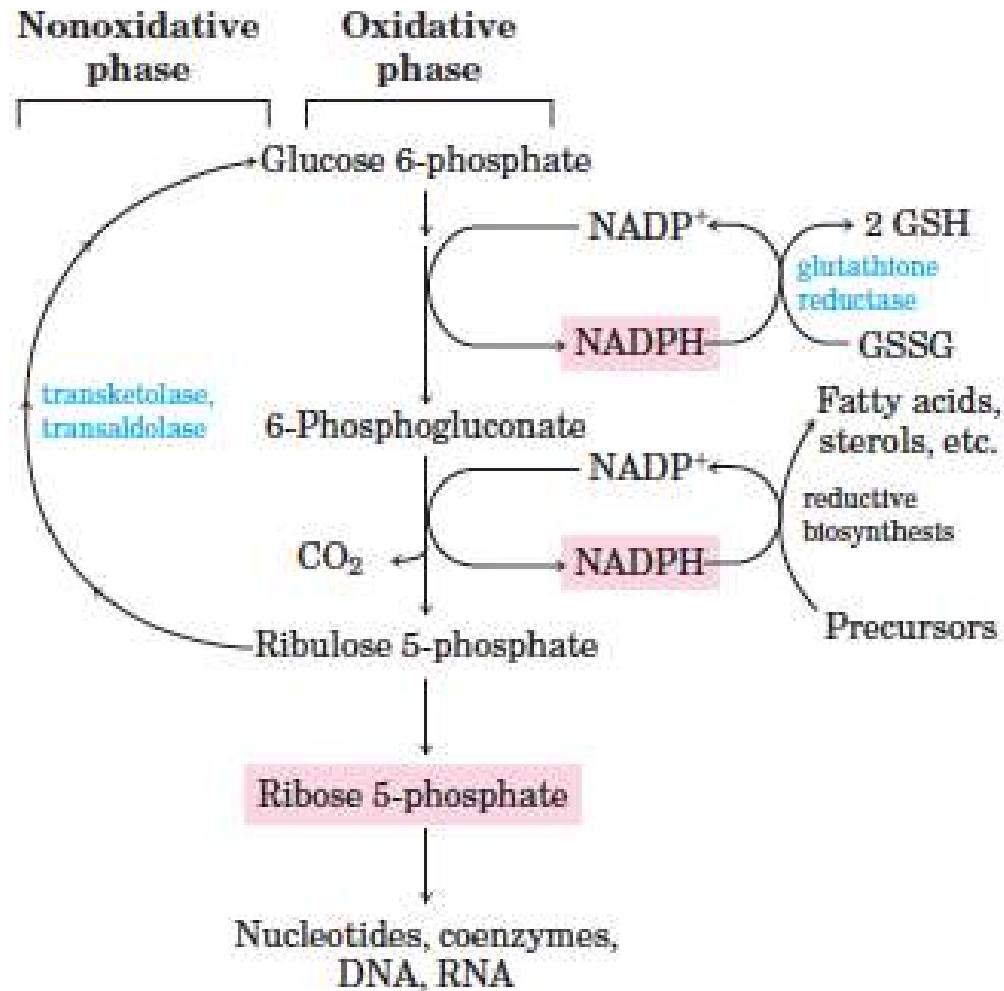
The pentose pathway is a shunt.

- Occurs in the cytosol.
- Begins with the glycolytic intermediate glucose 6-P.
- Reconnects with glycolysis:
 - Via two of the end products of this pathway i.e., glyceraldehyde 3-P and fructose 6-P.
 - These two intermediates further broken down in the glycolytic pathway.
 - This is why the pathway is referred as a shunt.

It's a shunt

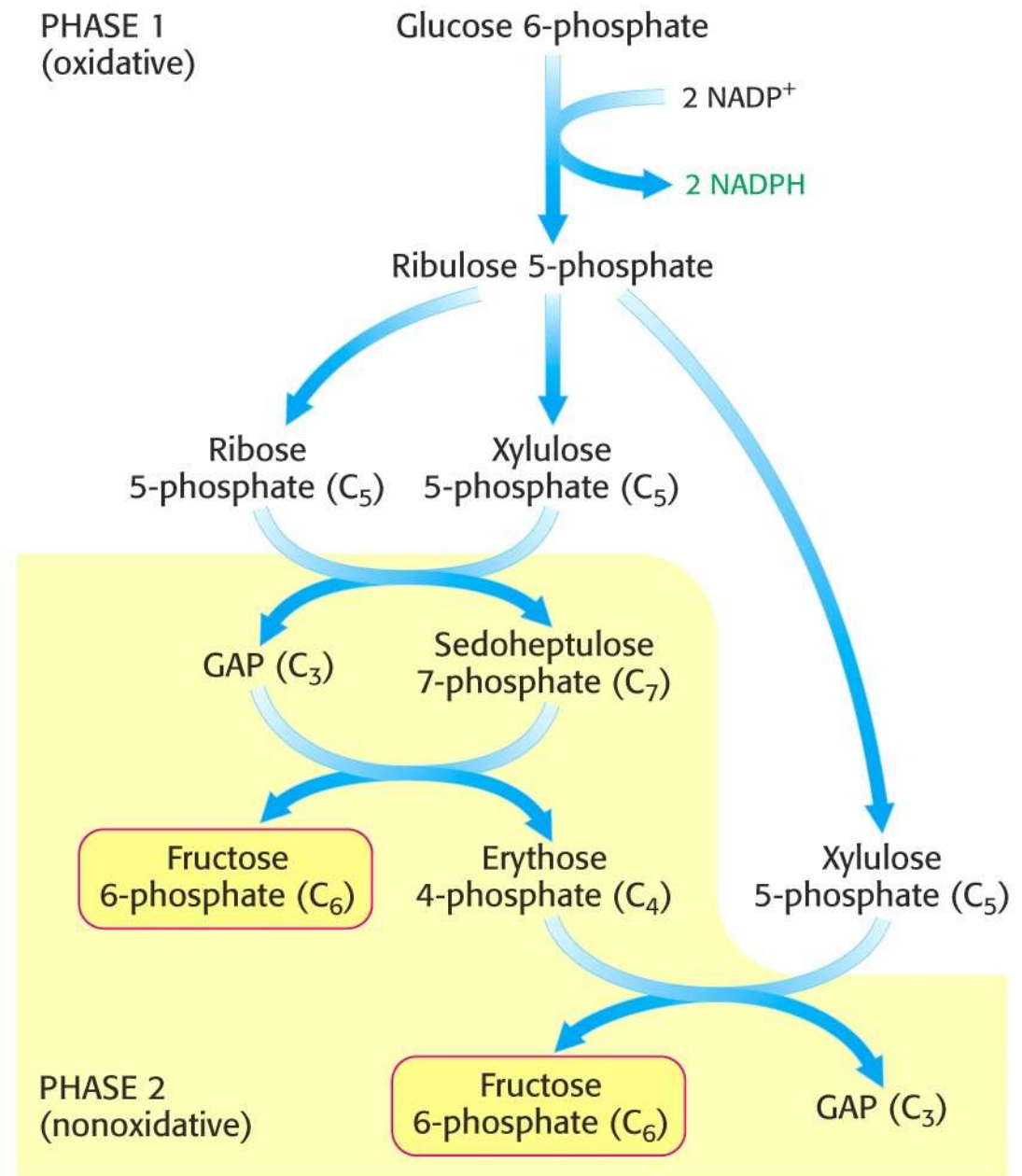


Overview of HMP shunt



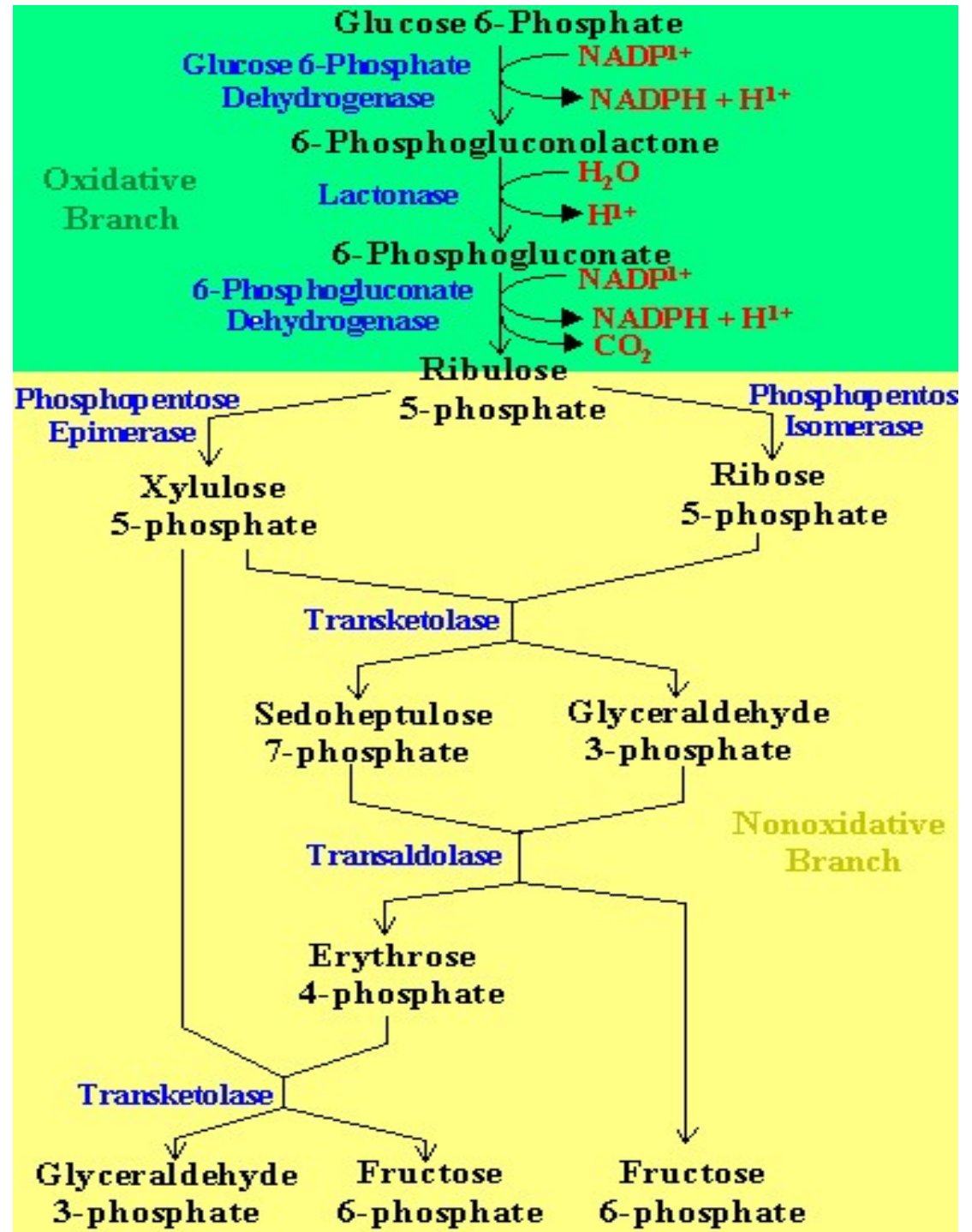
Overview of HMP shunt: Oxidative and Non-oxidative Phases

- Oxidative phases
 - Reactions producing NADPH
 - Irreversible
- Non-oxidative phases
 - Produces ribose-5-P
 - Reversible reactions feed to glycolysis



Outline of HMP shunt:

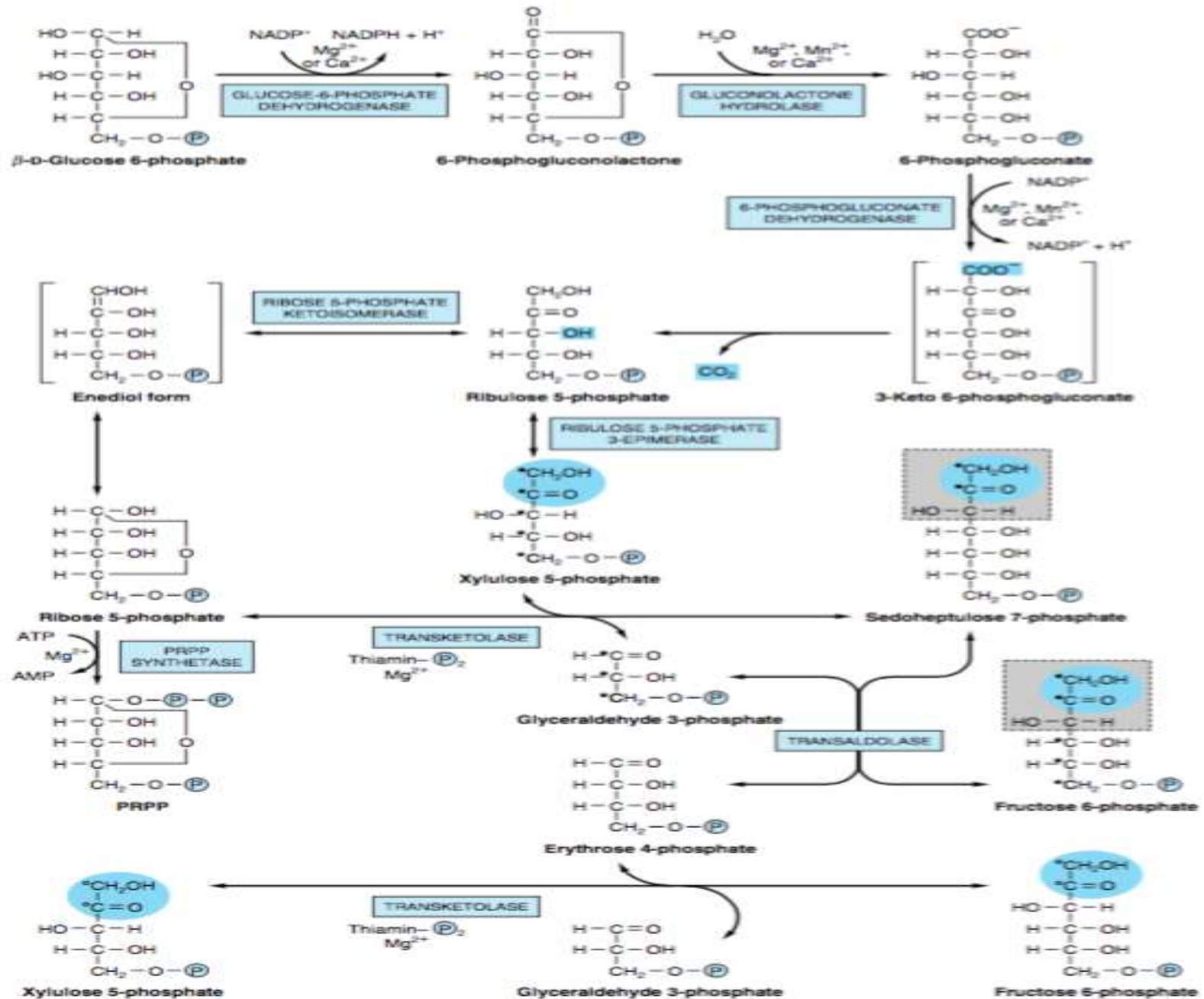
- NADPH + H⁺ is formed from two separate reactions.
- The glucose 6-phosphate DH (G6PD) reaction is the rate limiting step and is essentially irreversible.
- Cells have a greater need for NADPH than ribose 5-phosphate.



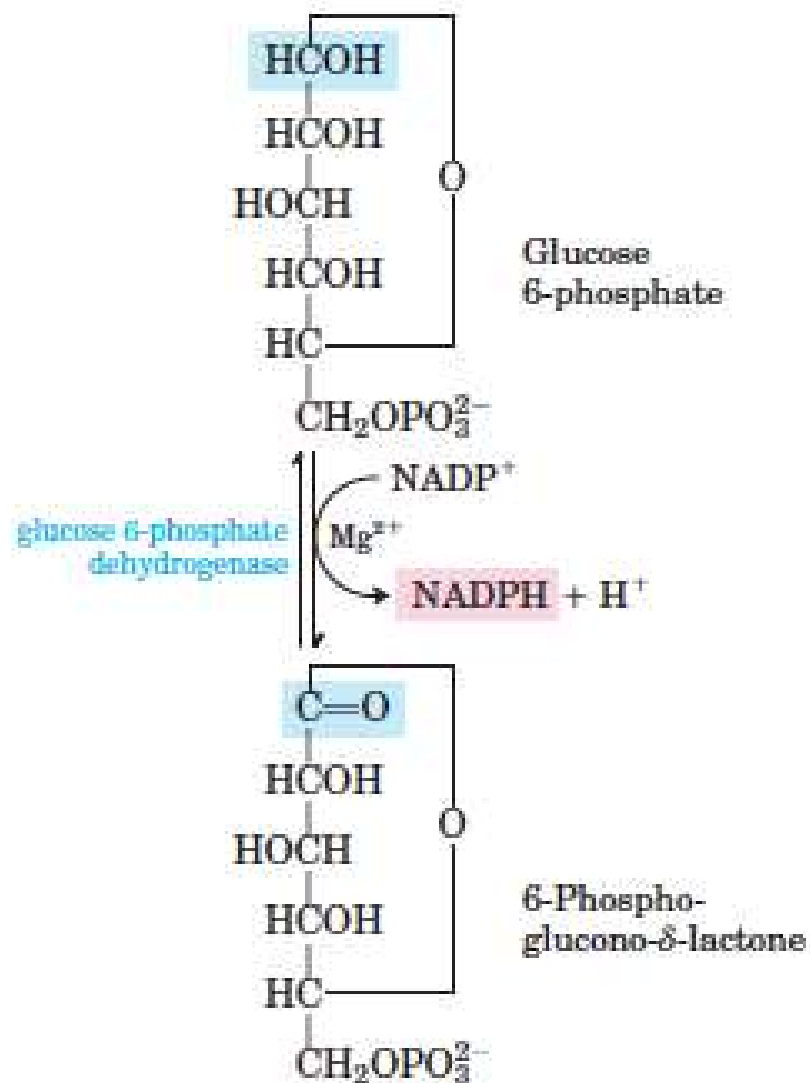
Reactions of HMP shunt

Dr Balakrishnan S, PhD.,

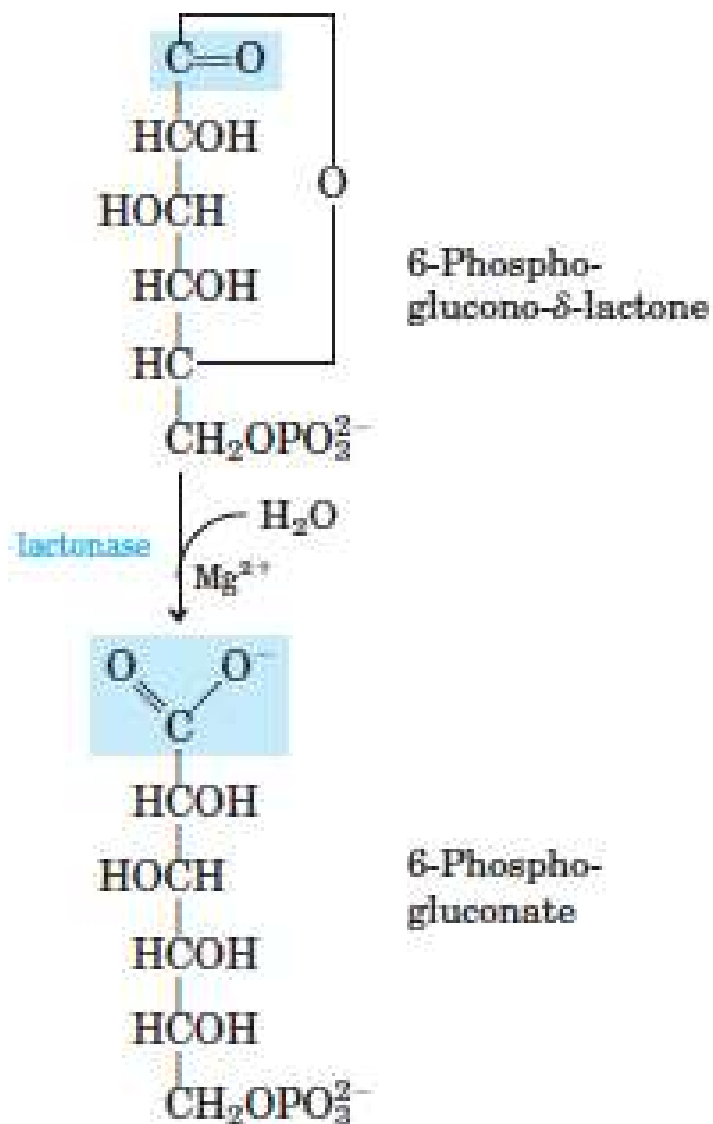
The pentose phosphate pathway



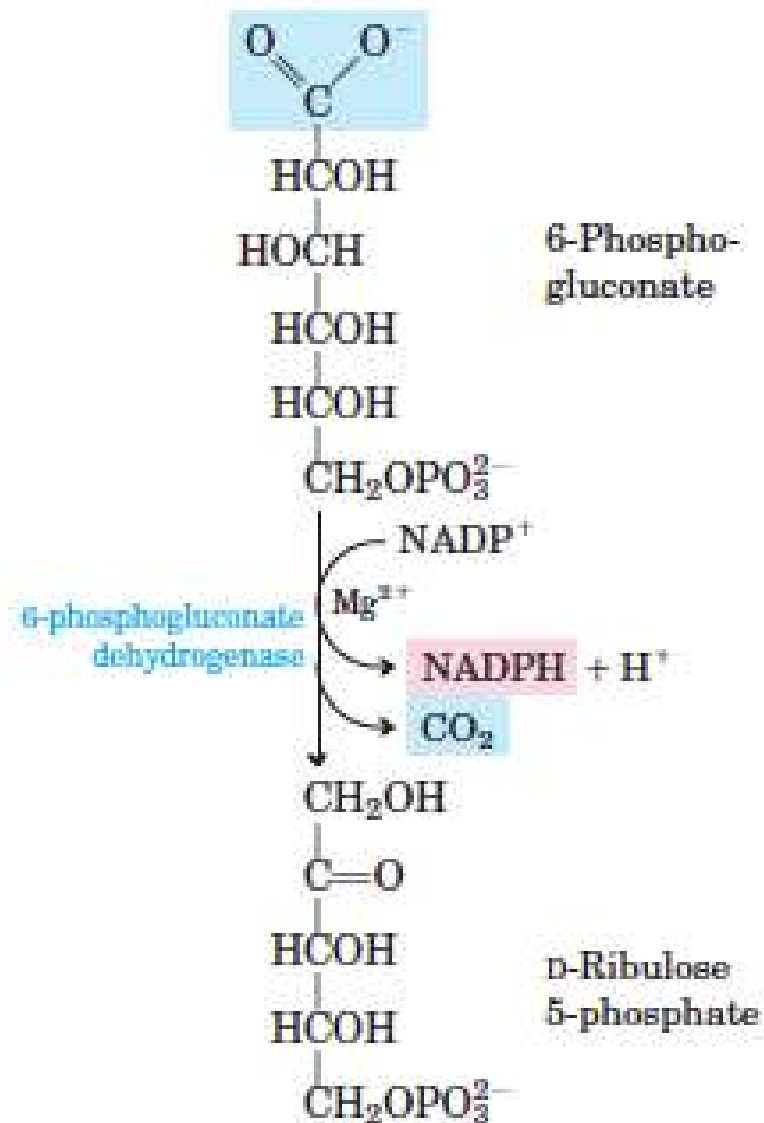
1. Glucose-6-phosphate to 6-Phosphoglucono- δ -lactone



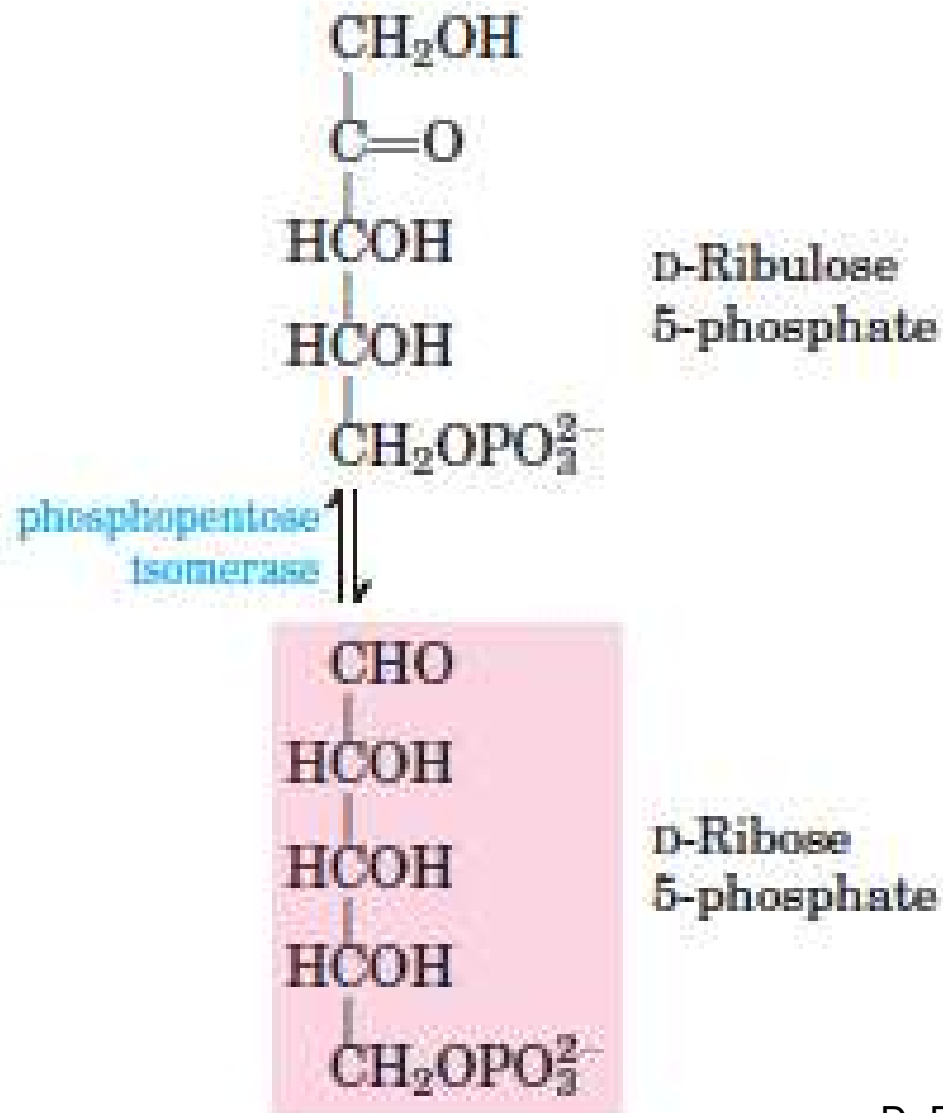
2. 6-Phosphoglucono- δ -lactone to 6-Phosphogluconate



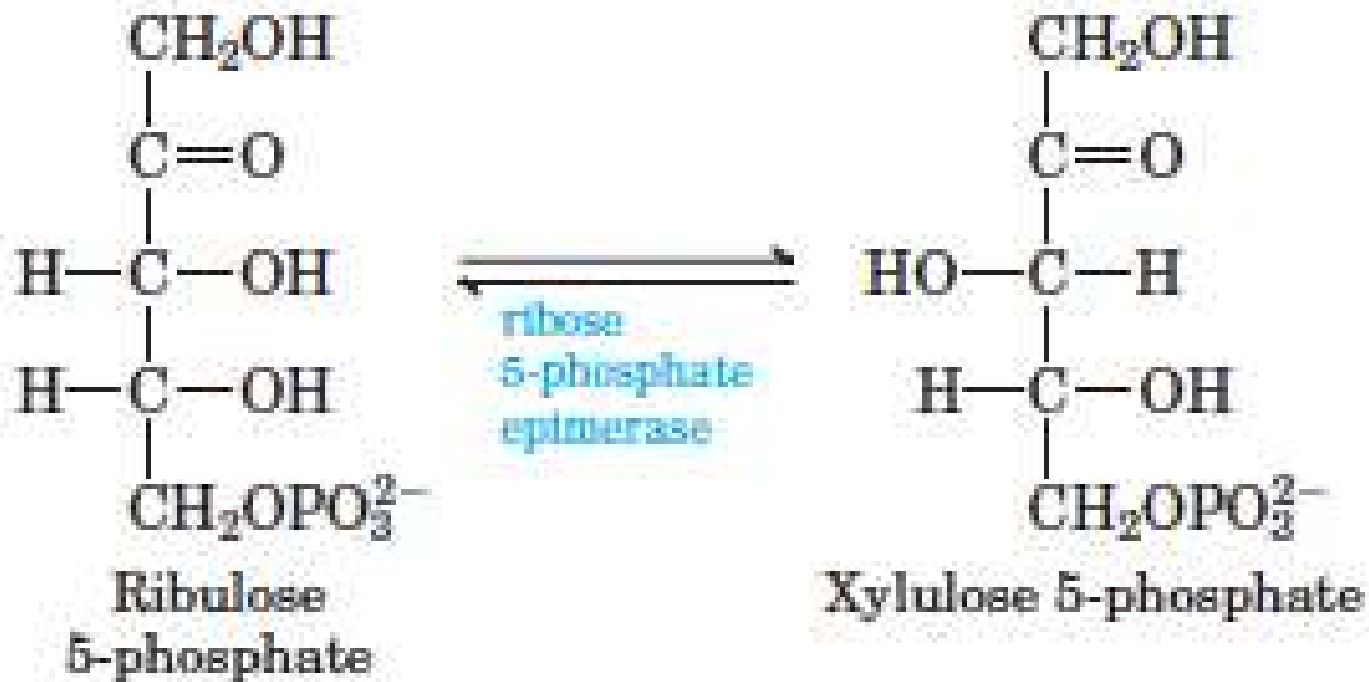
3. 6-Phosphogluconate to D-Ribulose 5-phosphate



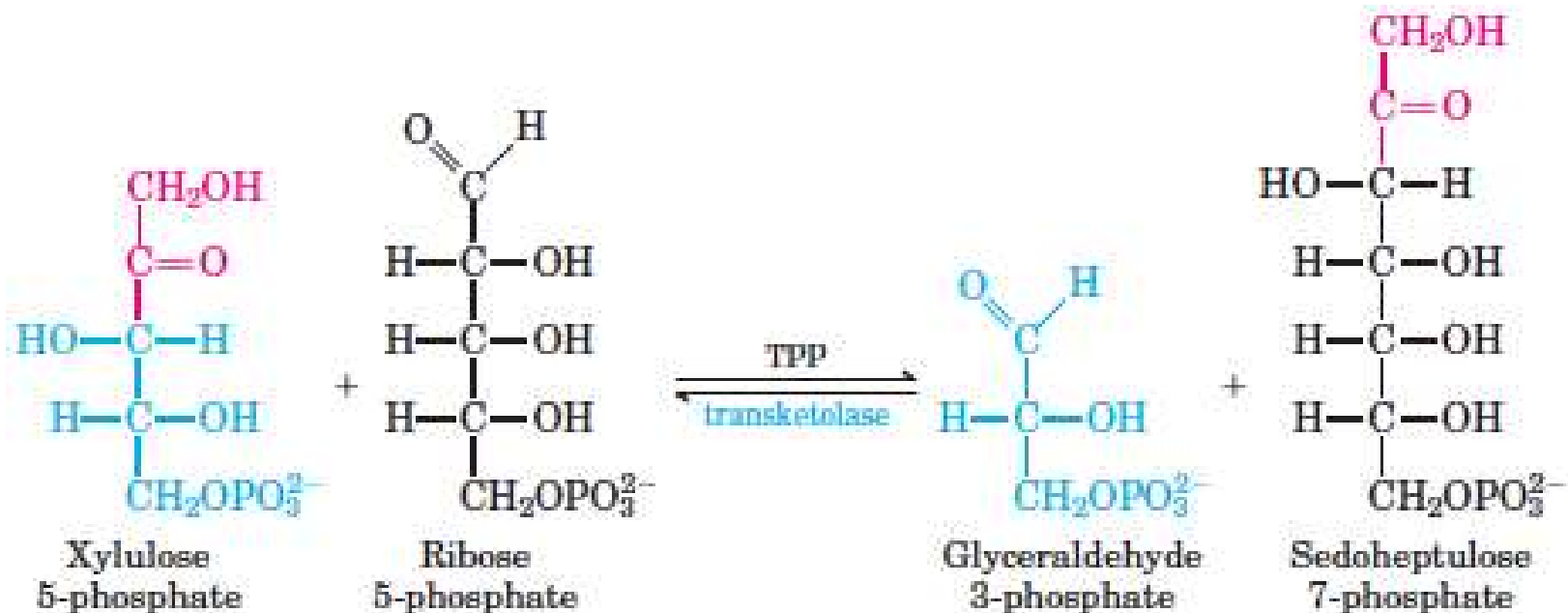
4. D-Ribulose 5-phosphate to D-Ribose 5-phosphate



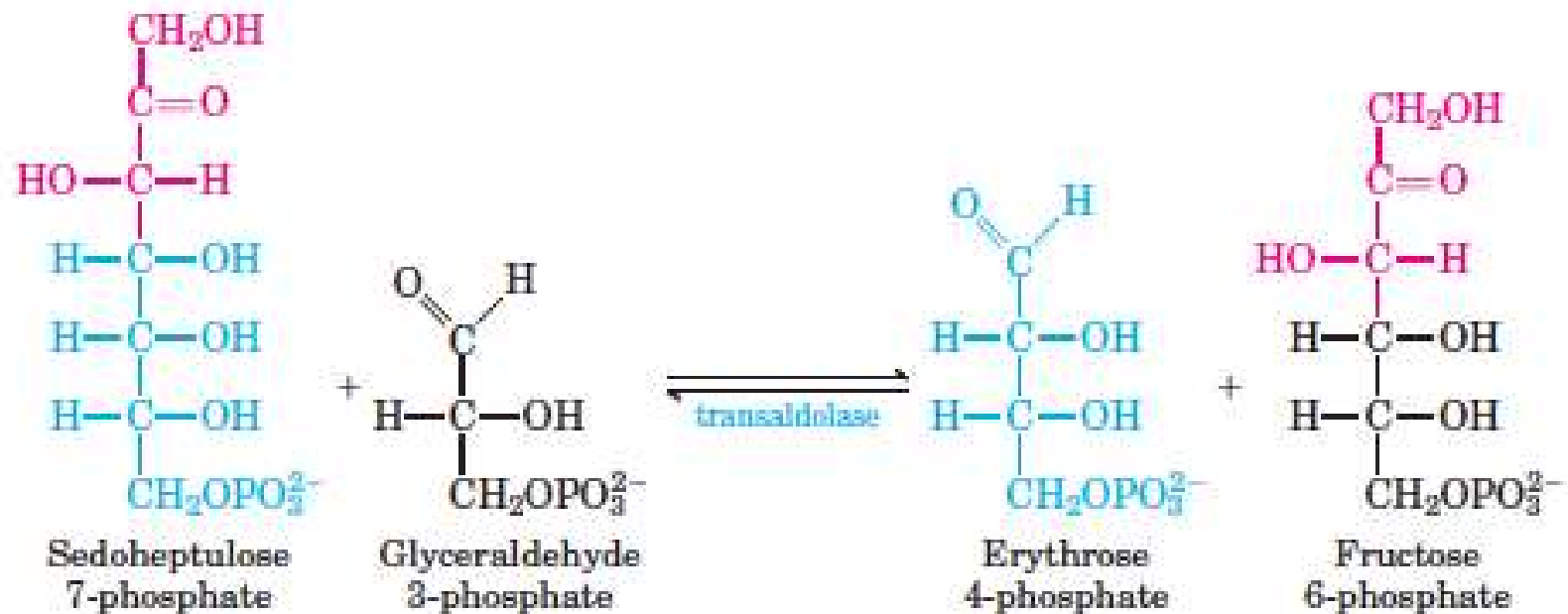
5. Ribulose 5-phosphate to xylulose 5-phosphate



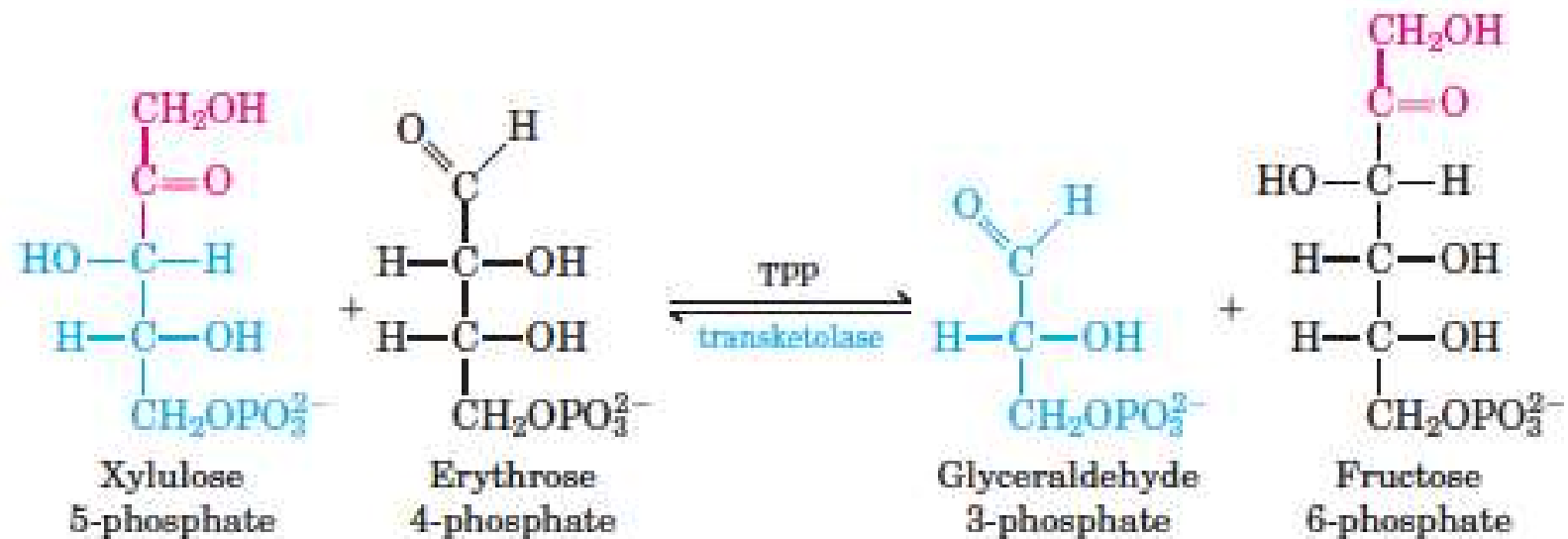
6. The first reaction catalyzed by Transketolase:

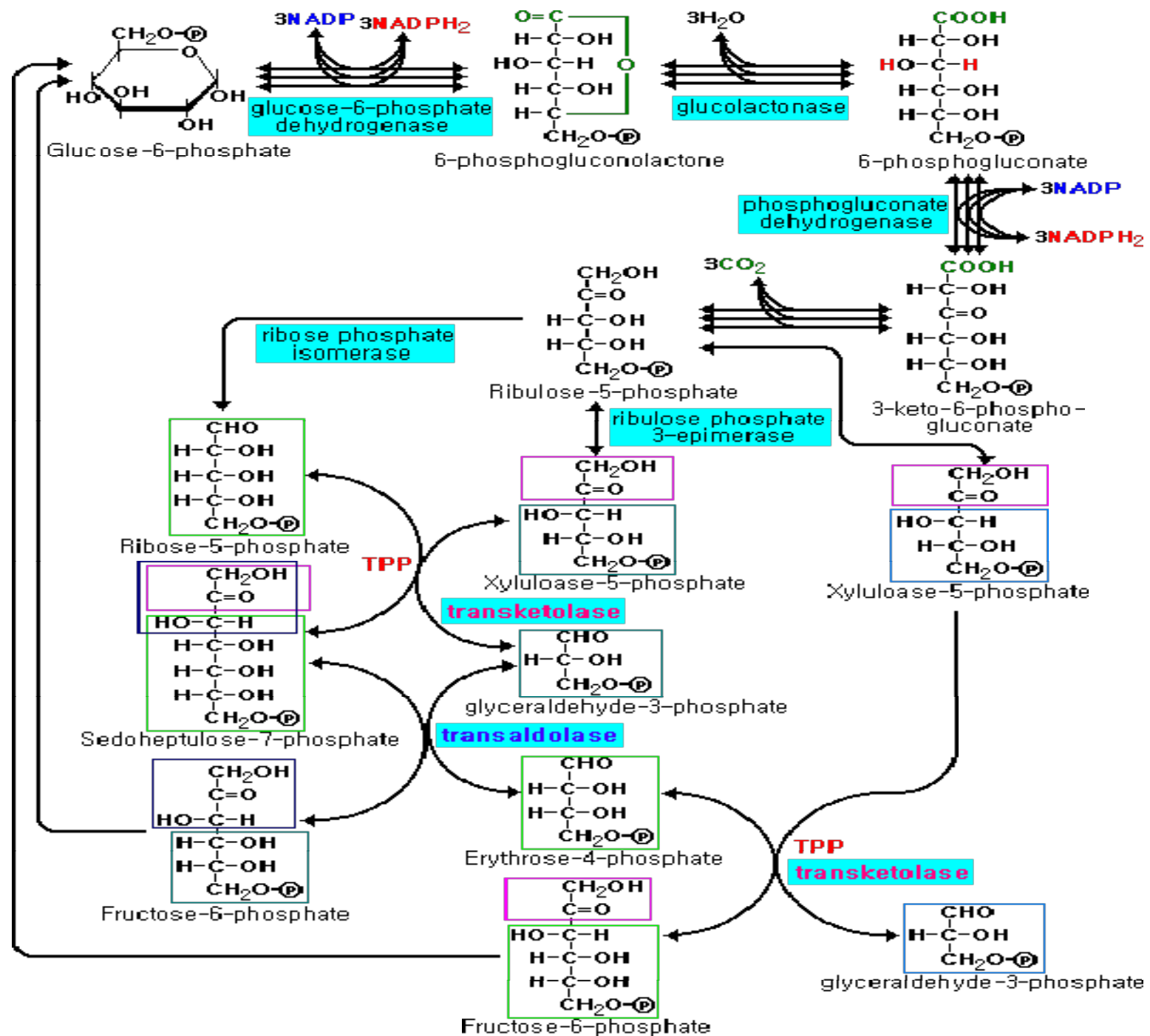


7. The reaction catalyzed by Transaldolase



8. The second reaction catalyzed by transketolase





Summary of pentose phosphate pathway

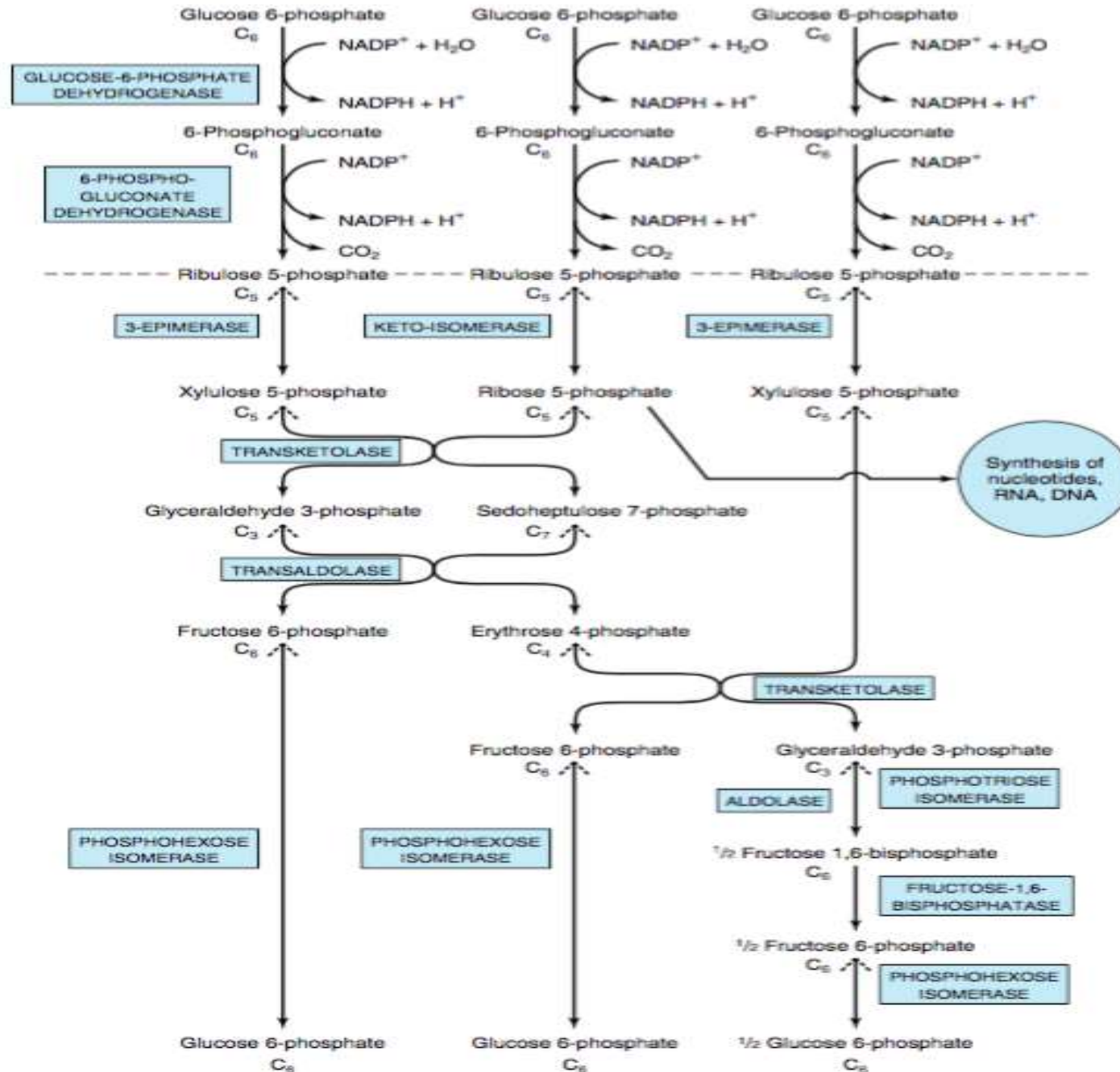


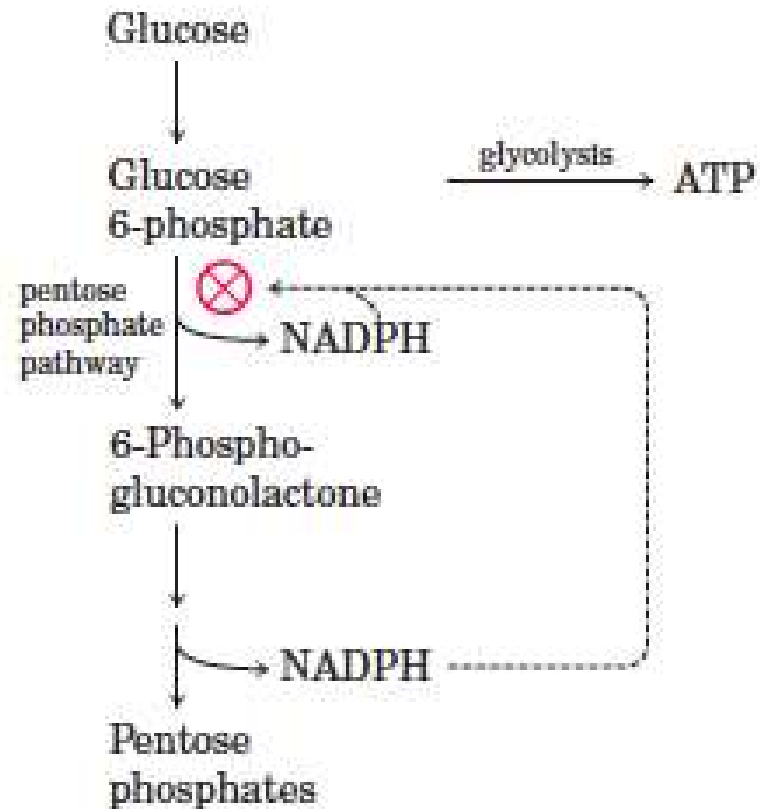
TABLE 20.3 Pentose phosphate pathway

Reaction	Enzyme
Oxidative phase	
Glucose 6-phosphate + NADP ⁺ $\longrightarrow$ 6-phosphoglucono- δ -lactone + NADPH + H ⁺	Glucose 6-phosphate dehydrogenase
6-Phosphoglucono- δ -lactone + H ₂ O $\longrightarrow$ 6-phosphogluconate + H ⁺	Lactonase
6-Phosphogluconate + NADP ⁺ $\longrightarrow$ ribulose 5-phosphate + CO ₂ + NADPH	6-Phosphogluconate dehydrogenase
Nonoxidative Phase	
Ribulose 5-phosphate $\rightleftharpoons$ ribose 5-phosphate	Phosphopentose isomerase
Ribulose 5-phosphate $\rightleftharpoons$ xylulose 5-phosphate	Phosphopentose epimerase
Xylulose 5-phosphate + ribose 5-phosphate $\rightleftharpoons$ sedoheptulose 7-phosphate + glyceraldehyde 3-phosphate	Transketolase
Sedoheptulose 7-phosphate + glyceraldehyde 3-phosphate $\rightleftharpoons$ fructose 6-phosphate + erythrose 4-phosphate	Transaldolase
Xylulose 5-phosphate + erythrose 4-phosphate $\rightleftharpoons$ fructose 6-phosphate + glyceraldehyde 3-phosphate	Transketolase

Regulation of the Pentose Pathway

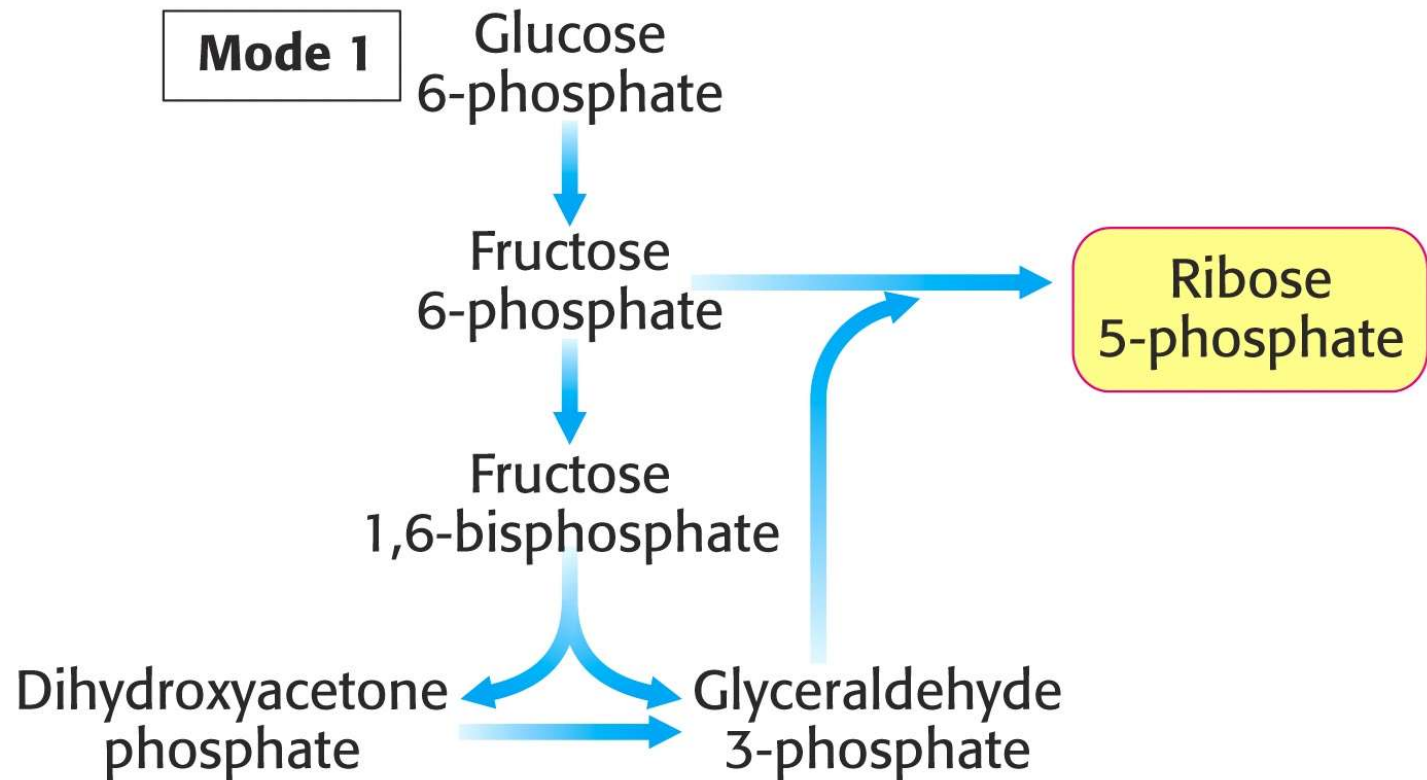
- Glucose 6-phosphate DH is the regulatory enzyme.
- NADPH is a potent competitive inhibitor of the enzyme.
 - Usually the ratio $\text{NADPH}/\text{NADP}^+$ is high so the enzyme is inhibited.
 - But, with increased demand for NADPH, the ratio decreases and enzyme activity is stimulated.
- The reactions of the non-oxidative portion of the pentose pathway are readily reversible.
 - The concentrations of the products and reactants can shift depending on the metabolic needs of a particular cell or tissue.

Glucose 6-Phosphate Is Partitioned between Glycolysis and the Pentose Phosphate Pathway

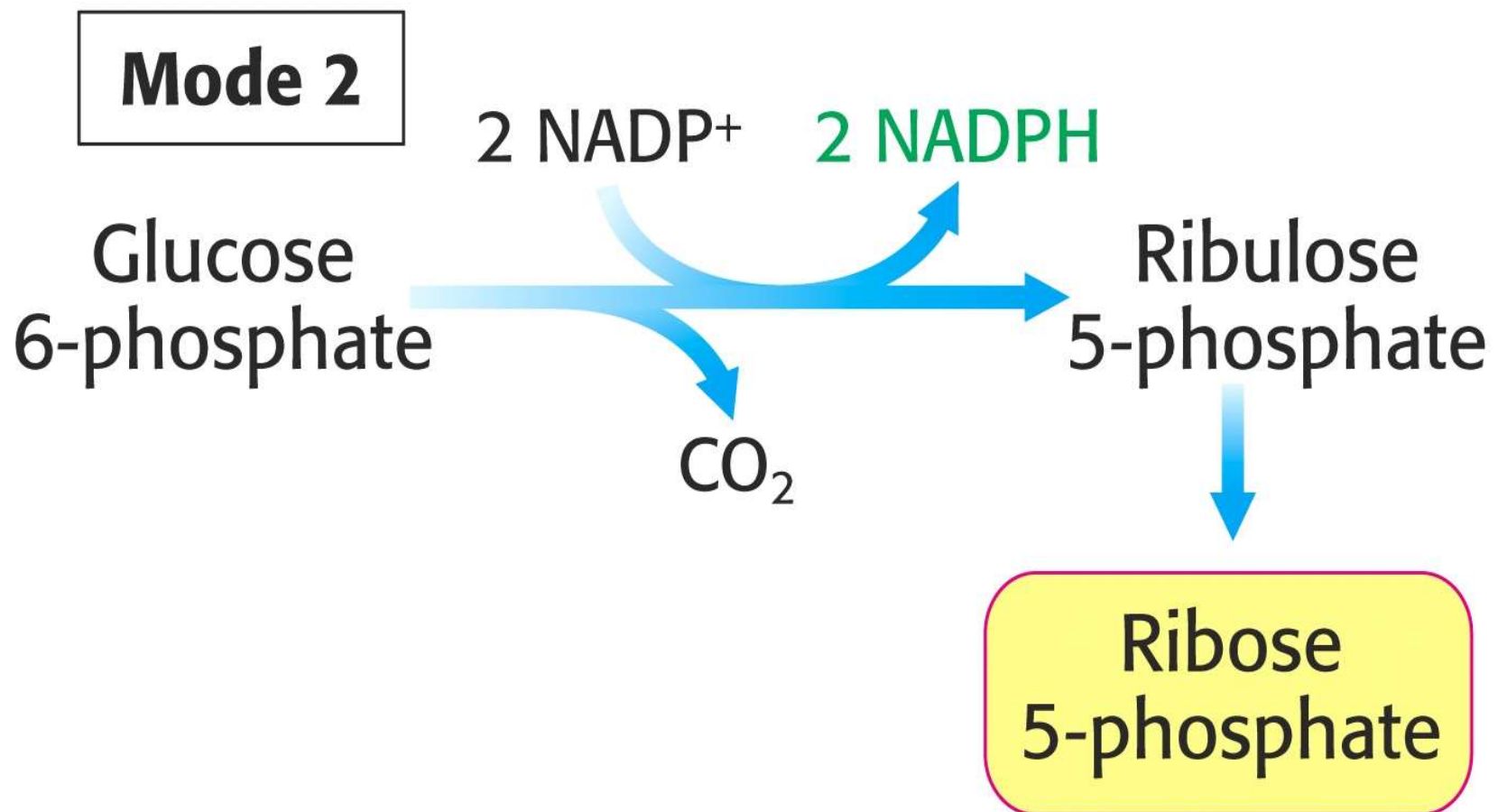


When NADPH is forming faster than it is being used for biosynthesis and glutathione reduction, NADPH rises and inhibits the first enzyme in the pentose phosphate pathway. As a result, more glucose 6-phosphate is available for glycolysis.

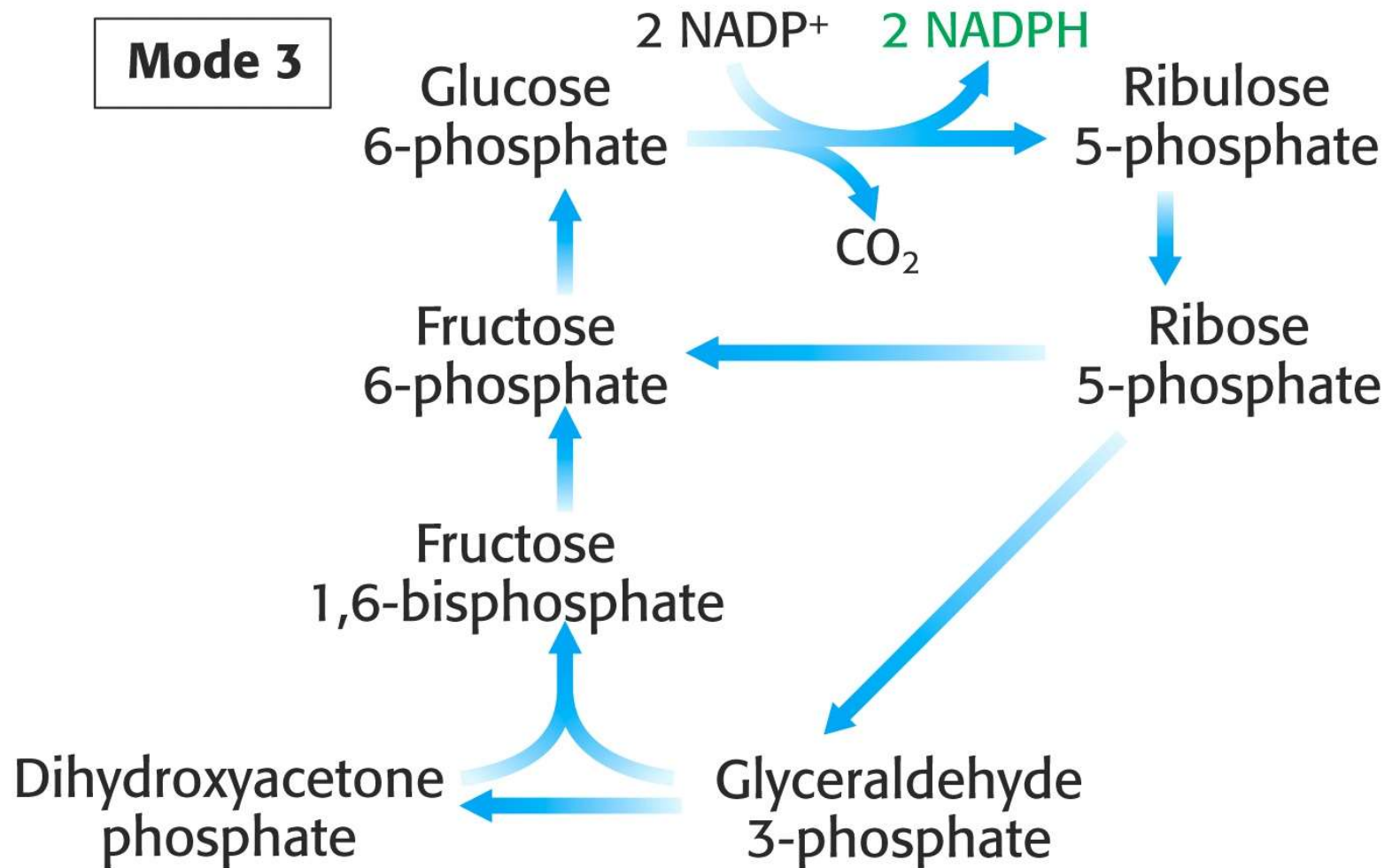
Rapidly dividing cells require more ribose 5-phosphate than NADPH.



The need for NADPH and ribose 5-phosphate is balanced.



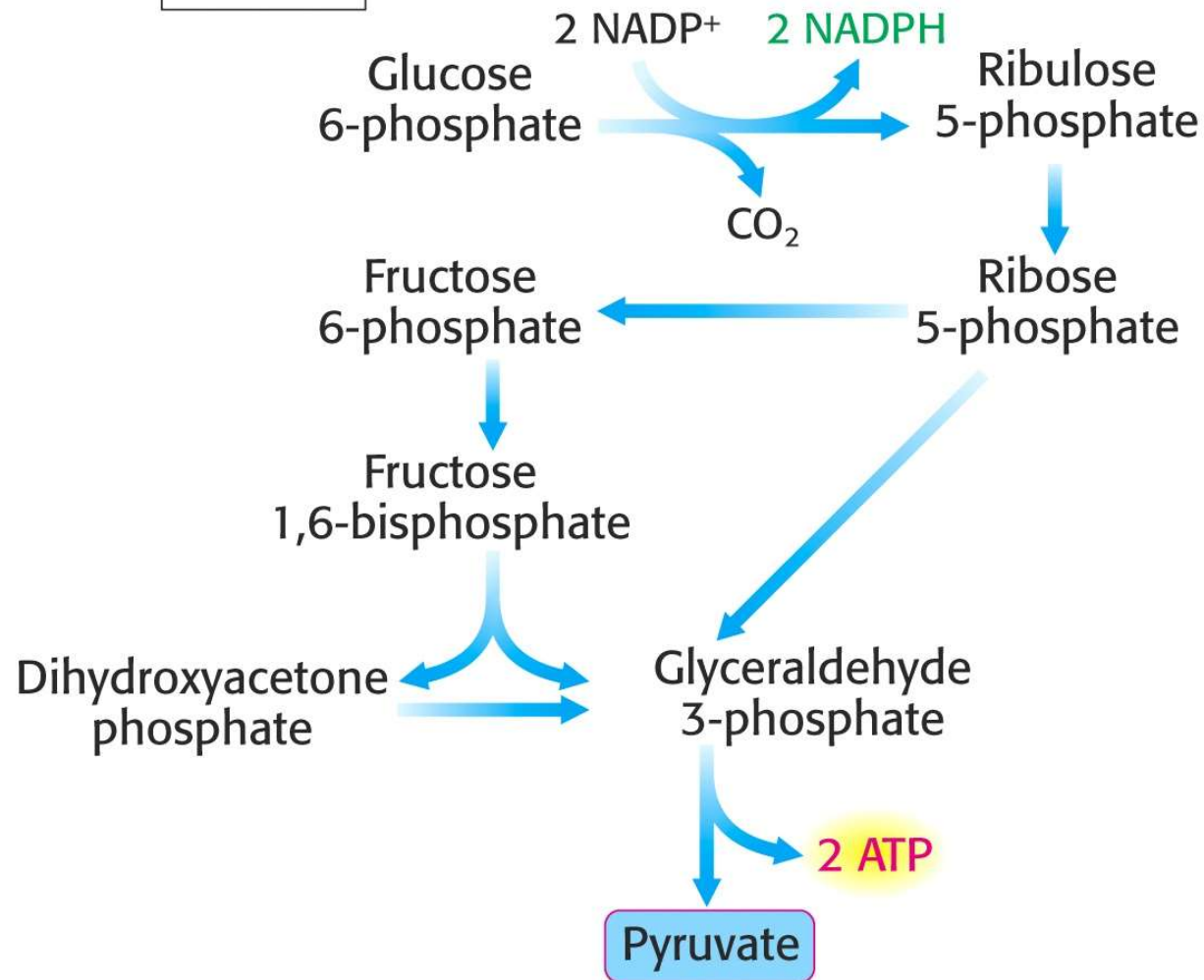
More NADPH is needed than ribose 5-phosphate;
Fatty acid synthesis in adipose cells.



Dr Balakrishnan S, PhD.,

The cell needs both NADPH and ATP

Mode 4



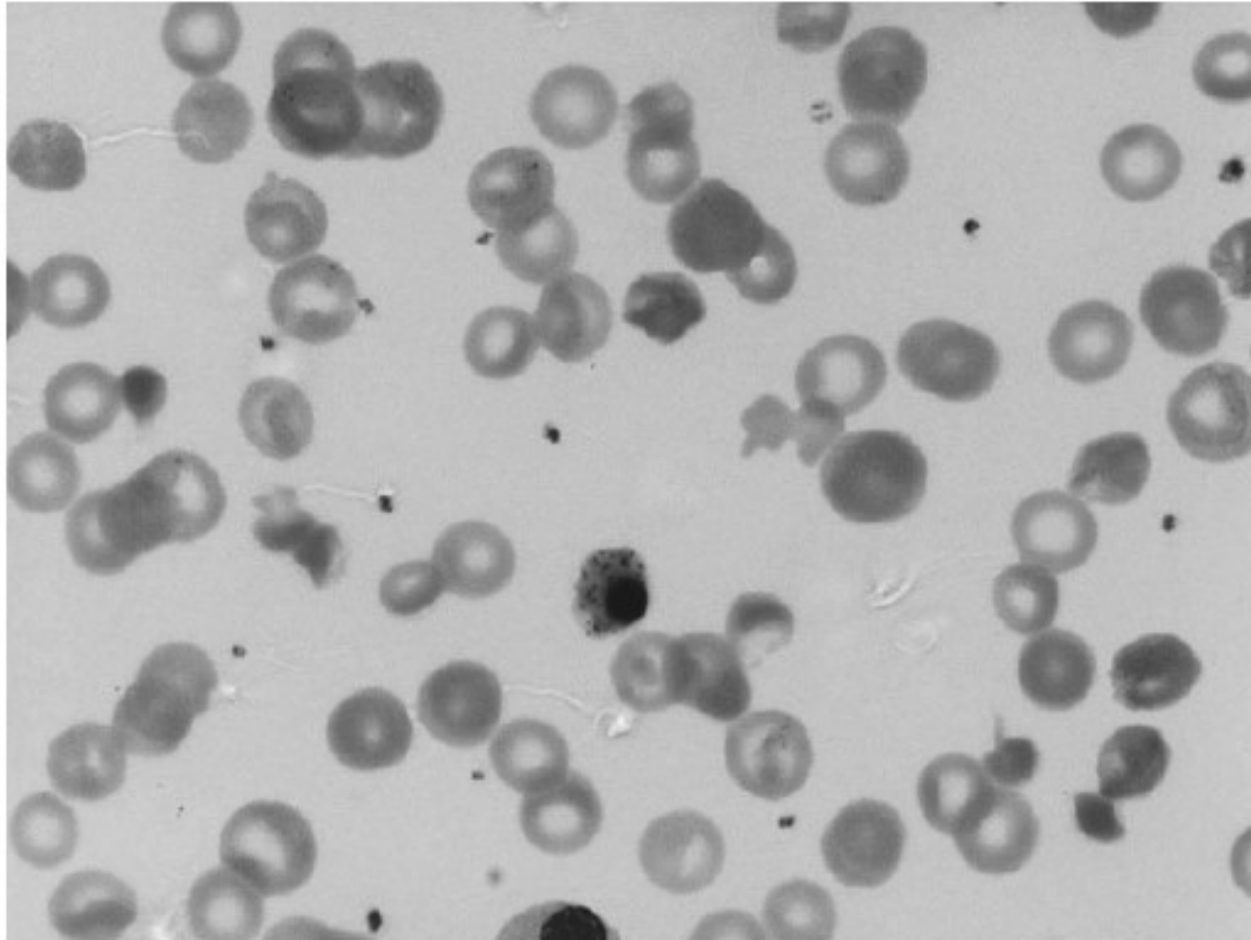
Difference of EM Pathway and HMP Shunt

<i>EM Pathway</i>	<i>HM Pathway</i>
1. Occurs in all tissues 2. Not a multicyclic process 3. Oxidation by dehydrogenation but NAD^+ is H-acceptor 4. ATP is required and ATP is produced 5. CO_2 is never formed	1. Occurs in certain special tissues for special function 2. Multicyclic process 3. Oxidation achieved by dehydrogenation but NADP^+ is used as H-acceptor 4. Not meant for energy; ATP is not produced, ATP is required for glucose to glucose-6-P. (for phosphorylation) and for interconversion of Pentoses 5. CO_2 is produced

Glucose 6-Phosphate Dehydrogenase Deficiency

- Result in hemolytic anemia
- Over 400 different mutations in the G6PDH gene (Xq28) are known.
- The major symptom is either an acute episodic or (rarely) a chronic hemolysis.
- The disease is X-linked recessive.
 - Males more severely affected
 - Female heterozygous for G6PDH deficiency have increased resistance to malaria.
 - The growth *Plasmodium falciparum* (malaria parasite) fails in G6PD deficient individuals.
 - Consequently, the deficiency is seen more commonly in families from regions where malaria is endemic.
- Hemolytic crises occur upon exposure to drugs (Primaquine) or diet components (fava beans) that cause enhanced formation of ROS.

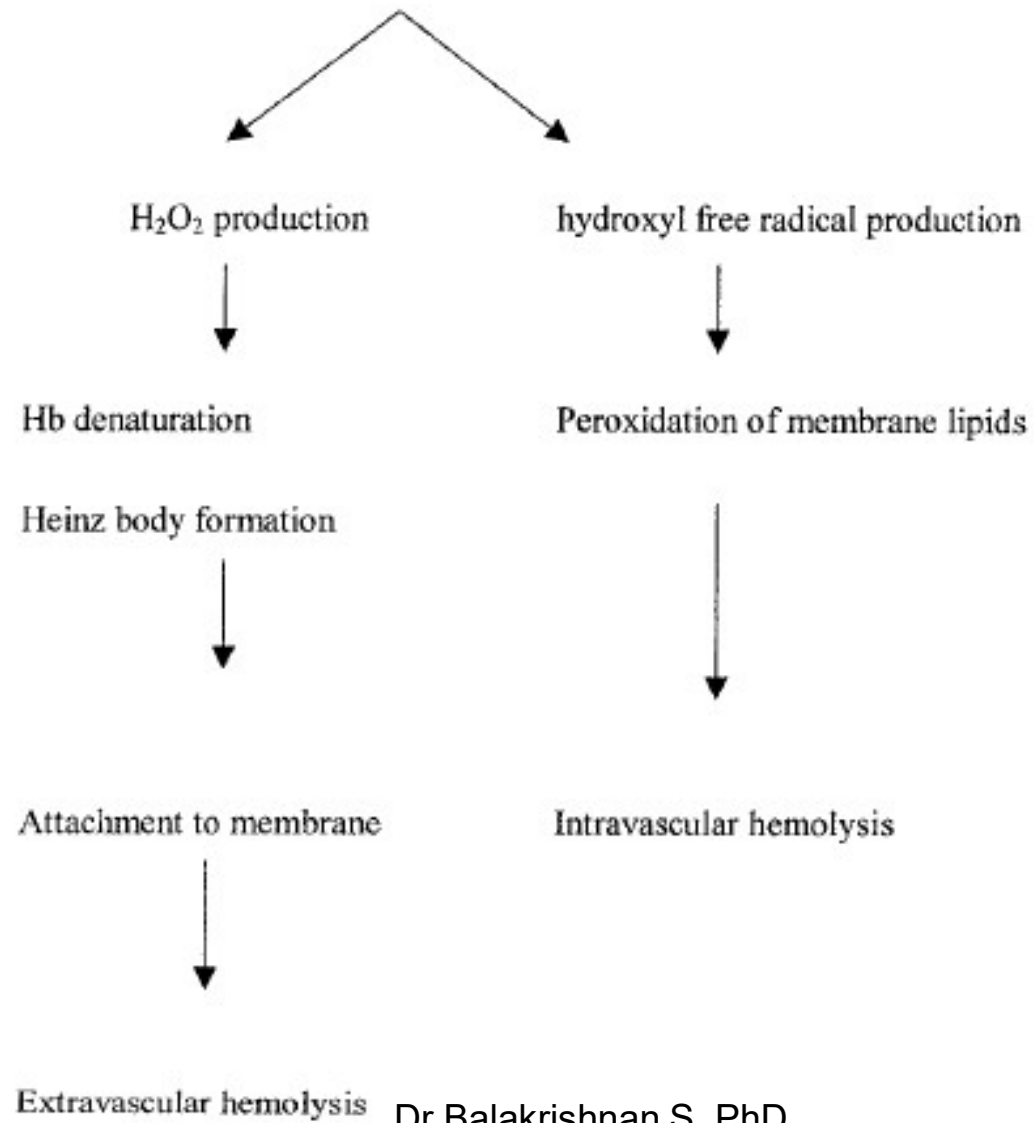
Peripheral blood features of oxidative hemolysis in G6PDHase Deficiency



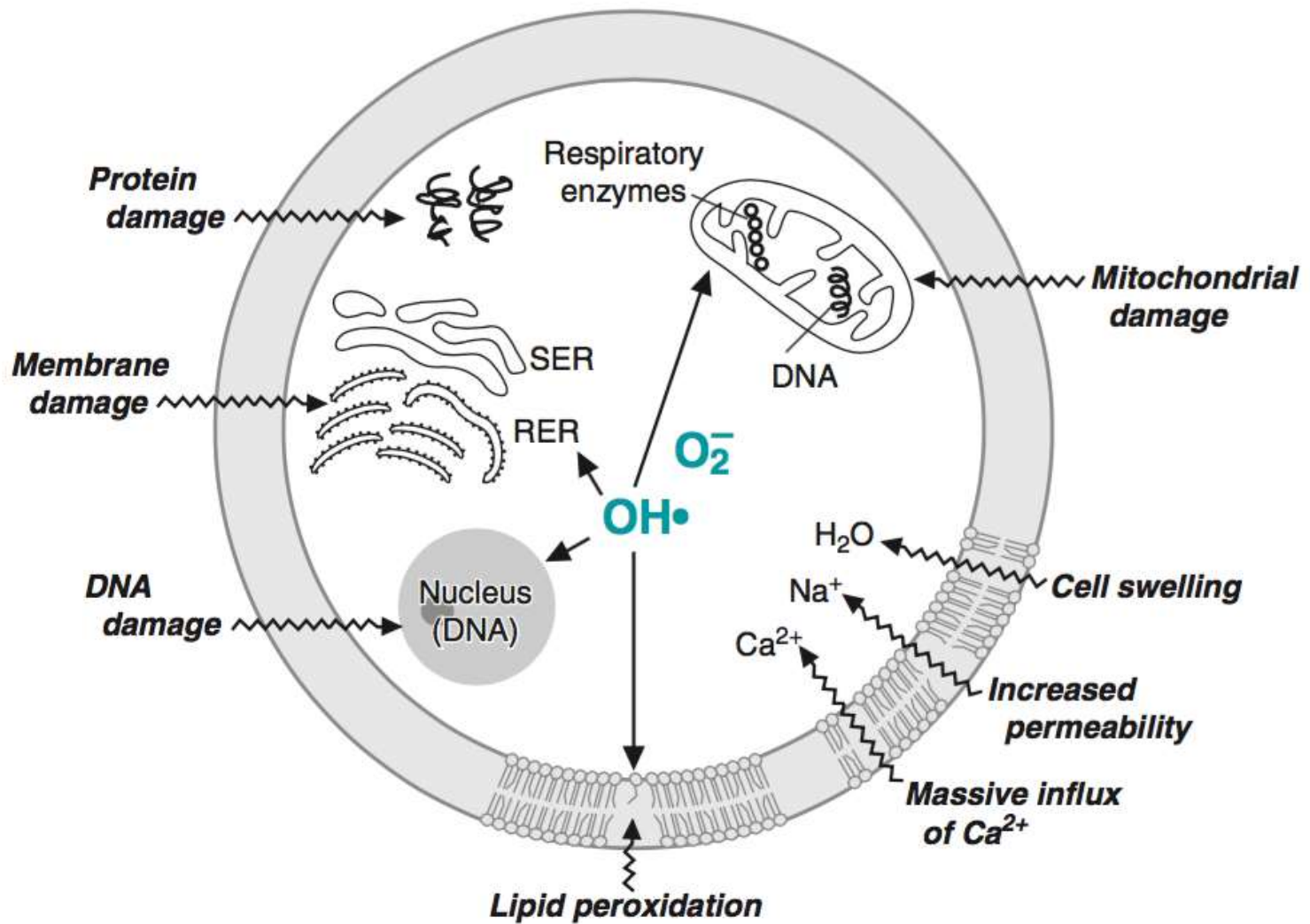
Dr Balakrishnan S, PhD.,

Mechanism of hemolysis in G6PDHase deficiency

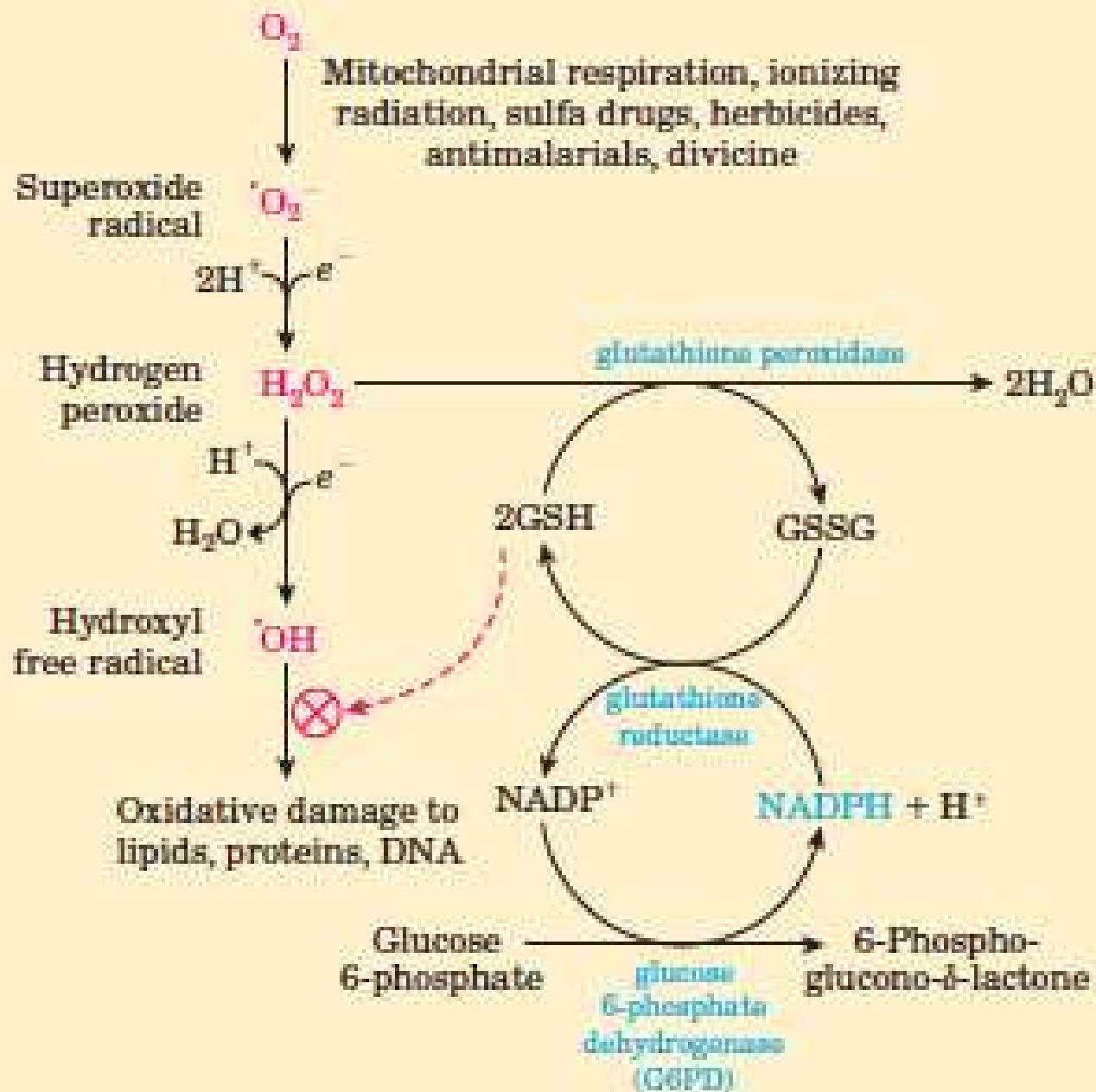
Free radical formation secondary to drugs/chemicals



Free radical-induced cellular injury

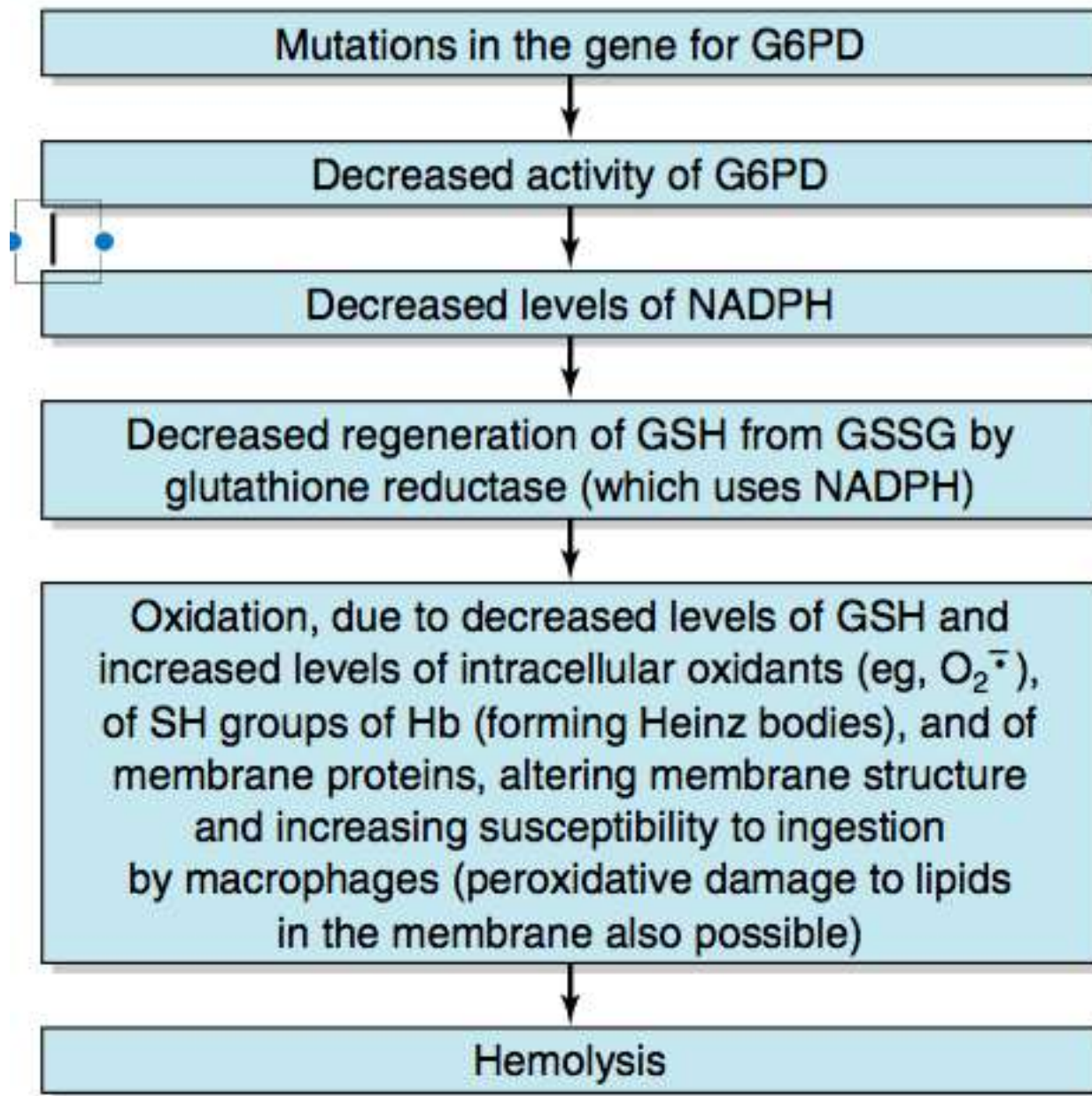


Dr Balakrishnan S, PhD.,



Case file: G6PD deficiency

- A patient with Glucose 6 phosphate dehydrogenase (G6PD) deficiency presents with symptoms of hemolytic anemia, including pale skin, jaundice, dark-colored urine, fever, weakness, and shortness of breath, following exposure to certain triggers such as infections and certain medications.
- Diagnosis:
 - a simple blood test,
 - G6PD activity assay or
 - genetic testing for G6PD mutations.
- Treatment:
 - prevention of exposure to triggers, as aspirin, certain antibiotics and antimalarial medications, fava beans, and mothballs
 - treating any triggering infection, and
 - checking with healthcare providers before taking any medication.



FAVISM

- Individuals with G6PD deficiency must not eat Fava beans.
- Hemolytic crises occur upon exposure to fava beans that cause enhanced formation of ROS - is called favism.
- Most likely because **the beans contain potential oxidants**
 - The pyrimidine aglycones of glucosides i.e., Divicine (2,6-diamino-4,5dihydropyrimidine) and Isouramil (5,6-dihydroxycytosine) are the oxidant implicated in favism.
 - Degree of hemolysis correlates to the hepatic metabolism of fava bean oxidants as well as the amount of divicine or isouramil in each fava bean and how the fava beans are consumed.
- Erythrocytes lyse=dark or black urine.
- Preceding symptoms:
 - Headache, nausea, back pain, chills, and fever are followed by hemoglobinuria and jaundice.
- The fall in hemoglobin is acute and severe and can be fatal if blood transfusion is not performed.

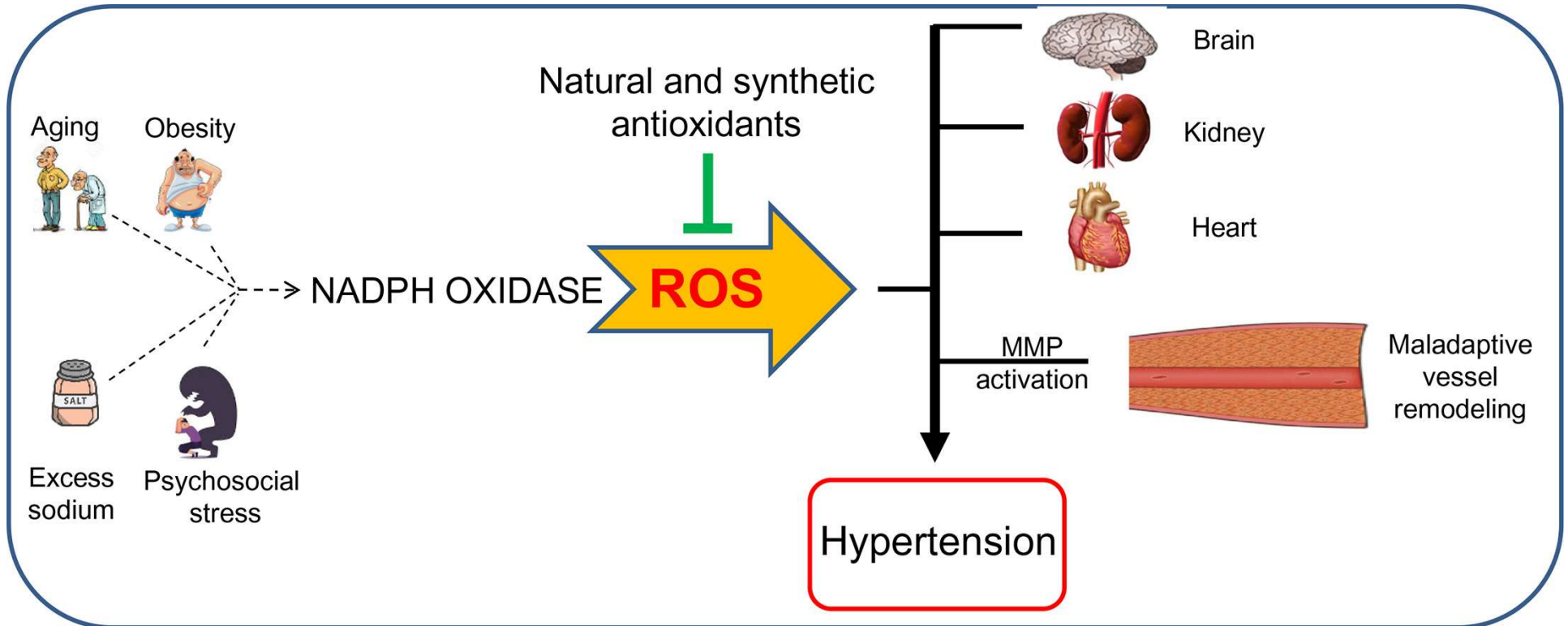
Case file: Favism

- A patient with favism presents with symptoms of hemolytic anemia, including fatigue, weakness, and jaundice, after consuming fava beans.
- Biochemical changes:
 - G6PDH deficiency,
 - Oxidative stress, leading to hemolysis
 - Anemia.
- Diagnosis:
 - G6PDH activity assay or
 - Genetic testing for G6PD mutations.
- Management:
 - Avoid fava beans and certain medications,
 - Monitoring for complications such as hemolysis and anemia.
 - In severe cases, blood transfusions may be necessary.

Glucose-6-phosphate DHase deficiency and **Essential Hypertension**

- **Oxidative stress being a major contributory factor** to the development of essential hypertension (Rodrigo et al., 2011)
- **ROS can increase BP by stimulation of heart rate and force.** (Massaro et al., 2019)
 - ROS as mediators of vasoconstriction induced by angiotensin II, endothelin-1 and urotensin-II, among others.
- ROS may also contribute to chronic hypertension by inducing myocardial hypertrophy. (Massaro et al., 2019)
- **Oxidative stress may be as a result of G6PD deficiency:**
 - Since the reduction of hydrogen peroxide, free radicals and other forms of oxidative stress ultimately and heavily depend on G6PD for the generation of NADPH.
 - NADPH, a co-factor that participates in the regeneration of GSH.

ROS & Hypertension



- In essential hypertension, **oxidative stress may be associated with G6PD deficiency.**
 - Reduction of GSSG to GSH takes place in a reaction catalysed by glutathione reductase (Hecker et al., 2013).
 - Reduced GSH then, reduces H_2O_2 to H_2O , a reaction catalysed by glutathione peroxidase.
 - By this mechanism, the structure of red blood cells is maintained. There is also a balance between the reactive oxygen species (ROS) and antioxidants.
 - Thus, the oxidative stress is in minimal amounts and does not have adverse effects on the red blood cells and the development of essential hypertension (Rodrigo et al., 2008).

Transketolase deficiency

- Due to genetically **‘variant’ form of transketolase**, which cannot bind TPP, affecting the ‘transketolation’ reaction.
- The patient suffering from this disorder shows severe neuropsychiatric symptoms.
 - It is characterized by **lesions and haemorrhages near the III cerebral ventricles, deranged mental functions, depression, disorientation, loss of memory and mental confusions.**
- There may be abnormal gait or stance, and paralysis of eye movements.
- Assessment of erythrocytes **transketolase activity** is one of the diagnosis methods for thiamine deficiency.

Read more...



THANK YOU...